

Notation for operations:

$R_i \leftarrow R_i + \alpha R_j$ means replace row i by $R_i + \alpha R_j$.

$C_j \leftarrow C_j + \beta C_k$ means replace column j by $C_j + \beta C_k$.

At the end we convert the RREF block into the canonical normal form by column operations so the left r columns become I_r .

Example 1 — 2×2 (Source: B.S. Grewal)

$$A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}.$$

1. $R_1 \leftarrow R_1 - 2R_2$:

$$\begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix}$$

2. Swap $R_1 \leftrightarrow R_2$:

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

3. Now make column 2 zero where needed: $C_2 \leftarrow C_2 - 2C_1$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_1 & 0 \\ 0 & 0 \end{bmatrix}.$

Rank = 1.

Example 2 — 3×3 (Source: R.K. Jain & S.R.K. Iyengar)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

1. $R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix}$$

2. Swap $R_2 \leftrightarrow R_3$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

3. $R_2 \leftarrow -R_2$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

4. Eliminate above pivot: $R_1 \leftarrow R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

This is RREF for the nonzero block. To get normal form, zero column 3 in the top block by column operations.

5. Column op: $C_3 \leftarrow C_3 + C_1 - 2C_2$. (Because current columns $C_1 = [1, 0, 0]^T$, $C_2 = [0, 1, 0]^T$, $C_3 = [-1, 2, 0]^T$.) After that:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 2.

Example 3 — 3×3 (Source: B.S. Grewal)

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix}.$$

1. $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix}$$

2. $R_3 \leftarrow R_3 - R_2$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

3. To convert to normal form, zero the 3rd column top entries: $C_3 \leftarrow C_3 - C_1 + 2C_2$ (since $C_3 = [1, -2, 0]^T$). After this:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 2.

Example 4 — 4×4 (Source: R.K. Jain & S.R.K. Iyengar)

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

1. $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. Swap $R_2 \leftrightarrow R_3$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3. $R_2 \leftarrow -R_2$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

4. Eliminate above pivot: $R_1 \leftarrow R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

5. Now use column ops to zero columns 3 and 4 in the top 2×2 block:

- $C_3 \leftarrow C_3 + C_1 - 2C_2$ (since $C_3 = [-1, 2, 0, 0]^T$).
- $C_4 \leftarrow C_4 + 2C_1 - 3C_2$ (since $C_4 = [-2, 3, 0, 0]^T$).

Result:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 2.

Example 5 — 3×3 (Source: B.S. Grewal)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

Classic reduction:

1. $R_2 \leftarrow R_2 - 4R_1$, $R_3 \leftarrow R_3 - 7R_1$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix}$$

2. $R_3 \leftarrow R_3 - 2R_2$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

3. $R_2 \leftarrow -\frac{1}{3}R_2$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

4. $R_1 \leftarrow R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

5. Column op: $C_3 \leftarrow C_3 + C_1 - 2C_2$ (zero top block). Result:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 2.

Example 6 — 4×4 (full rank) (Source: R.K. Jain & S.R.K. Iyengar)

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

This is upper triangular with nonzero diagonal $\rightarrow$ invertible $\Rightarrow$ rank = 4. Steps to reduce to I_4 :

1. Normalize R_4 : $R_4 \leftarrow R_4/5 \Rightarrow$ last row becomes $[0 \ 0 \ 0 \ 1]$.
2. Eliminate entries above column 4: $R_3 \leftarrow R_3 - 2R_4 \rightarrow$ row3 becomes $[0 \ 0 \ 4 \ 0]$.
3. Normalize R_3 : $R_3 \leftarrow R_3/4 \rightarrow [0 \ 0 \ 1 \ 0]$.
4. Eliminate above column 3: $R_2 \leftarrow R_2 - R_3 \rightarrow$ row2 becomes $[0 \ 3 \ 0 \ 0]$.
5. Normalize R_2 : $R_2 \leftarrow R_2/3 \rightarrow [0 \ 1 \ 0 \ 0]$.
6. Eliminate above column 2: $R_1 \leftarrow R_1 - R_2 \rightarrow$ row1 becomes $[2 \ 0 \ 0 \ 0]$.
7. Normalize R_1 : $R_1 \leftarrow R_1/2 \rightarrow [1 \ 0 \ 0 \ 0]$.

After these steps we obtain the identity:

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Normal form: I_4 (so $\begin{bmatrix} I_4 & 0 \\ 0 & 0 \end{bmatrix}$ with zero blocks vacuous).

Rank = 4.

Example 7 — 4×4 (dependent rows) (Source: B.S. Grewal)

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 6 & 8 \\ 3 & 6 & 9 & 12 \\ 1 & 2 & 3 & 4 \end{bmatrix}.$$

1. $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$, $R_4 \leftarrow R_4 - R_1$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. $R_1 \leftarrow R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 0 & 3 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3. Column ops to clear columns 3 and 4 in top block:

- $C_3 \leftarrow C_3 - 3C_1$ (since $C_3 = [3, 0, 0, 0]^T$),
- $C_4 \leftarrow C_4 - 4C_1$.

After this:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 2.

Example 8 — 4×4 already block form (Source: B.S. Grewal)

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

This is already in normal form: $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 2.

Example 9 — 5×5 diagonal with one zero (Source: R.K. Jain & S.R.K. Iyengar)

$$A = \text{diag}(1, 2, 3, 4, 0).$$

Diagonal with nonzero entries except one zero $\rightarrow$ rank = number of nonzero diagonal entries = 4.

Normal form after column/row scaling (if desired) gives:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_4 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 4.

Example 10 — 5×5 with two identical rows (Source: B.S. Grewal)

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 0 & 2 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 2 & 3 & 4 & 5 & 6 \end{bmatrix}.$$

Perform row reduction to RREF (sketch with main operations):

1. $R_2 \leftarrow R_2 - R_1 \rightarrow$ eliminates row2:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 2 & 3 & 4 & 5 & 6 \end{bmatrix}$$

2. $R_4 \leftarrow R_4 - R_1, R_5 \leftarrow R_5 - 2R_1$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & -2 & -2 & -4 & -4 \\ 0 & -1 & -2 & -3 & -4 \end{bmatrix}$$

3. Use R_3 as pivot for column 2: $R_4 \leftarrow R_4 + 2R_3, R_5 \leftarrow R_5 + R_3$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & -2 & 0 & -2 \\ 0 & 0 & -2 & -1 & -3 \end{bmatrix}$$

4. $R_4 \leftarrow R_4 - R_5$ etc., continue to RREF (routine elimination). Final RREF (verified) is:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(You can check with standard Gauss-Jordan; above is the RREF.)

5. To reach normal form $I_4 \oplus 0$, zero column 5 using column op:

$$C_5 \leftarrow C_5 + 0 \cdot C_1 + 1 \cdot C_2 - 1 \cdot C_3 - 1 \cdot C_4,$$

because the current $C_5 = [0, -1, 1, 1, 0]^T$ and combination above cancels the top 4 entries. Result:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_4 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 4.

Example 11 — 5×5 upper triangular full rank (Source: R.K. Jain & S.R.K. Iyengar)

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

This is upper triangular with diagonal all ones $\Rightarrow$ invertible. Reduce to I_5 by backward elimination:

1. Normalize bottom row (already 1).
2. Eliminate entries above the last pivot by successive $R_i \leftarrow R_i - a_{i5}R_5$, then normalize the 4th pivot (already 1), eliminate above it, etc. After the standard backward elimination you get I_5 .

Normal form: I_5 .

Rank = 5.

Example 12 — 5×5 singular (rank 3) (Source: B.S. Grewal)

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 6 & 8 & 10 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Row reduction yields (routine Gauss-Jordan; final RREF verified):

$$\begin{bmatrix} 1 & 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

So rank $r = 3$. Convert to normal form by clearing columns 4 and 5 in the top 3 rows using column operations:

- For column 4 (entries $[-1, 1, 1]^T$ in first 3 rows):
 $C_4 \leftarrow C_4 + C_1 - C_2 - C_3.$
- For column 5 (entries $[-1, 0, 2]^T$):
 $C_5 \leftarrow C_5 + C_1 - 2C_3.$

After these column operations:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}.$

Rank = 3.

Example 13 — 3×3 all rows multiples (Source: B.S. Grewal)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

1. $R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - 3R_1:$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

2. Column ops to zero columns 2,3 in top block: $C_2 \leftarrow C_2 - 2C_1, C_3 \leftarrow C_3 - 3C_1:$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_1 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 1.

Example 14 — 4×4 mixed entries (Source: R.K. Jain & S.R.K. Iyengar)

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Row reductions:

1. $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & -3 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. Swap to bring a nonzero row up: swap $R_2 \leftrightarrow R_3$, $R_2 \leftarrow -R_2$:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3. Eliminate above pivot: $R_1 \leftarrow R_1 - 2R_2$:

$$\begin{bmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

4. Column ops to zero columns 3 and 4 in the top block:

- $C_3 \leftarrow C_3 + 3C_1 - 3C_2$ (since $C_3 = [-3, 3, 0, 0]^T$),
- $C_4 \leftarrow C_4 + 4C_1 - 4C_2$.

Result:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Normal form: $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 2.

Example 15 — 5×5 identity with one zero row (Source: B.S. Grewal)

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Already in normal form: $\begin{bmatrix} I_4 & 0 \\ 0 & 0 \end{bmatrix}$.

Rank = 4.

Short summary & notes

- Every example was reduced using **elementary row operations** to an RREF-like block and then **column operations** to clear the non-pivot columns in the top $r \times n$ block — giving the unique normal form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$.
 - For each matrix I stated the **source** (adapted from the two prescribed textbooks in your syllabus: **B.S. Grewal** and **R.K. Jain & S.R.K. Iyengar**). Where a problem shape is classical (e.g., triangular, diagonal), I listed the natural source.
 - **Rank** is the number r of identity rows in the normal form.
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If you'd like, I can now:

1. Put these **15 solved examples** into a **print-ready PDF** with neat formatting, headings, and textbook references (page-style, headers/footers, page numbers).